



**PSGR
Krishnammal College for Women**



INTEGRATING OPENAI WITH UIPATH FOR SMART INSURANCE PROCESSING

PROJECT WORK SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE OF BACHELOR OF COMPUTER SCIENCE

Submitted By

Harishmitha S V

21BCC011

Guided By

Dr. M. Sowmya, M.Sc., MBA., M.Phil., Ph.D.,

Assistant Professor,

Department of Computer Science.

DEPARTMENT OF COMPUTER SCIENCE

MARCH 2024

CERTIFICATE

This is to certify that the project work entitled “**INTEGRATING OPENAI WITH UIPATH FOR SMART INSURANCE PROCESSING**” submitted to Bharathiar University, Coimbatore for the award of Degree of the Bachelor of Computer Science with Cognitive Systems is a record of the original work done by **Harishmitha S V (21BCC011)**, under the supervision and guidance of **Dr. M. Sowmya**, Assistant Professor, Department of Computer Science, PSGR Krishnammal College for Women, Coimbatore and this project work has not formed the basis for the award of any Degree/ Diploma/ Associate/ Fellowship or similar title to any candidate of any university.

Forwarded by

Dr. M. Sowmya

Faculty Guide

Dr. S. Karpagavalli

Head of the Department

Dr. P. Meena

Principal

Submitted for the final examination held on **Date**

Internal examiner

External examiner

DECLARATION

I hereby declare that this project work entitled “**INTEGRATING OPENAI WITH UIPATH FOR SMART INSURANCE PROCESSING**” submitted to Bharathiar University, Coimbatore for award of the Degree of Bachelor of Computer Science with Cognitive Systems is a record of the original work done by **Harishmitha S V (21BCC011)**, under the supervision and guidance of **Dr. M. Sowmya**, Assistant Professor, Department of Computer Science, PSGR Krishnammal College for Women, Coimbatore and this project work has not formed the basis for the award of any Degree/ Diploma/ Associate/ Fellowship or similar title to any candidate of any university.

Place: Coimbatore

Harishmitha S V

Date:

(21BCC011)

Endorsed by

Place: Coimbatore

Dr. M. Sowmya

Date:

Faculty Guide

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“Success is to be measured not so much by the position that one has reached in life but as by the obstacle which he had to overcome while trying to succeed”.

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SYNOPSIS

The "Smart Insurance Processing" project utilizes UiPath and OpenAI to streamline the insurance processing workflow. The project employs AI algorithms to extract relevant details from personal certificates, which are then stored in an Excel sheet for easy access. The insurance form is automatically filled using the extracted data from the Excel sheet, reducing manual data entry and minimizing errors.

Furthermore, the project integrates a conversational AI system powered by OpenAI to handle user queries efficiently. Users are prompted to input their queries, which are then processed by the AI. Based on the user's preference, responses to the queries can be delivered via messages, emails, or phone calls, enhancing user experience and providing quick and convenient support.

In summary, the project enhances insurance processing by leveraging AI technologies to automate data extraction, form filling, and query handling, ultimately improving efficiency, accuracy, and user satisfaction in the insurance workflow.

1. INTRODUCTION

In an era defined by digital transformation, the fusion of artificial intelligence and robotic process automation (RPA) has become increasingly instrumental in streamlining and enhancing various aspects of business operations. One remarkable manifestation of this synergy is evident in the innovative project on smart insurance processing, leveraging the power of UiPath and OpenAI technologies. This project represents a paradigm shift in the insurance industry, revolutionizing the traditional processes associated with data retrieval, form filling, and customer query resolution.

At its core, the project harnesses the capabilities of AI to autonomously extract pertinent details from personal certificates, eliminating the tedious and error-prone manual data entry tasks. By seamlessly integrating UiPath's automation prowess, this solution ensures swift and accurate extraction of relevant information, subsequently storing it in an Excel database for easy accessibility and reference. This automation not only expedites the insurance processing workflow but also minimizes the likelihood of human-induced errors, thereby enhancing overall efficiency and reliability.

Moreover, the project extends its transformative impact beyond mere data extraction by automating the completion of insurance forms. Leveraging the extracted data stored in Excel, the system dynamically populates the requisite fields in the insurance documents, significantly reducing the time and effort expended on paperwork. This intelligent automation not only enhances operational efficiency but also enhances the customer experience by expediting the insurance application process, thereby fostering customer satisfaction and loyalty.

Furthermore, the project incorporates a sophisticated AI-driven query resolution system, enabling seamless communication between the insurer and the insured. By leveraging OpenAI's advanced natural language processing capabilities, users can articulate their queries in plain language, prompting the system to provide accurate and timely responses. What sets this project apart is its adaptive nature, allowing users to choose their preferred mode of communication for receiving responses, be it through messages, emails, or even phone calls, thereby catering to diverse user preferences and ensuring a personalized experience. This comprehensive approach not only enhances customer engagement but also fosters trust and transparency within the insurance ecosystem, ultimately driving operational excellence and competitive advantage.

1.1 Organization Profile:

- **Organisation name:** Life Insurance Corporation of India (LIC)
- **Founded:** 1956
- **Headquarters:** Mumbai, Maharashtra, India
- **Chairperson:** Siddhartha Mohanty
- **Industry:** Insurance
- **Products:** Life insurance, annuities, pensions
- **Insurance advisor:** Mr. M. Senthilkumar

1.2 System Environment

Hardware requirement:

Processor: 11th Gen Intel(R) Core(TM) i5-1155G7 @ 2.50GHz 2.50 GHz

Installed RAM 8.00 GB (7.65 GB usable)

System type: 64-bit operating system, x64-based processor

Pen and touch: No pen or touch input is available for this display

Edition Windows 11 Home Single Language

Version: 22H2

OS build: 22621.3155

Software requirement:

UiPath orchestrator: Online Account

UiPath Studio: Version higher than 20

OpenAI API Key: Gpt-3.5 or Gpt-4

1.3 Software Features

1. Document Processing with UiPath:

- Utilize UiPath for intelligent document processing (IDP) to extract relevant information from personal certificates of the person.
- Implement OCR (Optical Character Recognition) capabilities to extract text from scanned documents.
- Use UiPath workflows to automate the extraction and processing of data from various document formats.

2. Integration with OpenAI:

- Integrate OpenAI for natural language processing (NLP) capabilities to understand user queries.
- Use OpenAI's language models to comprehend and generate responses to user queries related to insurance documents and policies.

3. Excel Integration:

- Store extracted personal information in Excel sheets for easy access and management.
- Utilize Excel automation within UiPath to read and write data to Excel sheets seamlessly.

4. Automatic Form Filling:

- Develop algorithms to automatically populate insurance forms with the extracted data from personal certificates stored in Excel.
- Implement error handling mechanisms to ensure accuracy and completeness of filled forms.

5. User Query Handling:

- Provide a user interface where users can input queries regarding insurance policies or documents.

- Utilize OpenAI to understand and respond to user queries effectively.
- Offer multiple modes of communication for responses such as messages, emails, or phone calls.

6. Multi-Mode Communication:

- Implement functionality to allow users to choose their preferred mode of communication for receiving responses (e.g., SMS, email, phone call).
- Integrate with appropriate APIs or services for sending messages, emails, and making phone calls.

7. Security and Compliance:

- Implement robust security measures to protect sensitive personal and insurance-related data.
- Ensure compliance with relevant regulations such as GDPR (General Data Protection Regulation) or HIPAA (Health Insurance Portability and Accountability Act) depending on the jurisdiction.

8. Scalability and Performance:

- Design the system to handle a large volume of documents and user queries efficiently.
- Optimize algorithms and workflows to minimize processing time and resource utilization.

9. Logging and Monitoring:

- Implement logging functionality to track user interactions, system errors, and performance metrics.
- Set up monitoring tools to ensure the system is running smoothly and to identify and address any issues promptly.

1.4 Activity packages used

1. **UiPath.Excel.Activities:** Used for interacting with Excel files to store and retrieve data, such as details of the person extracted from personal certificates.
2. **UiPath.AI.DocumentUnderstanding.Activities:** This package is part of the Document Understanding capabilities of UiPath. It would have been used to extract relevant information from insurance documents, such as forms, using OCR (Optical Character Recognition) and AI-based techniques.
3. **OpenAI.Activities:** Employed to interact with the OpenAI API, enabling your project to utilize natural language processing (NLP) capabilities to respond to user queries. This would have facilitated the generation of responses based on the input received from the user.
4. **Twilio.Activities:** Utilized for integrating Twilio's messaging and telephony capabilities into your automation process. This would have enabled sending messages (SMS) or making phone calls to users based on their preferred mode of communication, delivering answers to their queries.

In our smart insurance processing project, we utilized a suite of UiPath activity packages to automate key tasks efficiently. The UiPath.Excel.Activities package facilitated seamless management of data, allowing us to store and retrieve information from Excel files with ease. Additionally, we employed the UiPath.AI.DocumentUnderstanding.Activities package to extract essential details from insurance documents using advanced OCR and AI technologies.

Integrating OpenAI.Activities into our workflow enhanced our system's capabilities by enabling sophisticated natural language processing. This empowered our system to intelligently respond to user queries. Furthermore, with the Twilio.Activities package, we seamlessly integrated Twilio's messaging and telephony features, enabling communication with users through their preferred mode, whether via SMS or phone calls. By combining these activity packages, we developed a cohesive automation solution that streamlined insurance processing while providing a user-friendly interface for interaction.

2. SYSTEM STUDY AND ANALYSIS

2.1 Existing System

Prior to the implementation of the smart insurance processing system, insurance agents were burdened with labor-intensive manual tasks. They spent considerable time and effort transcribing details from personal certificates into the insurance portal. This manual data entry process was not only time-consuming but also prone to errors, leading to inefficiencies and potential inaccuracies in customer records.

Meanwhile, customers faced significant challenges in accessing and understanding their insurance information. They were often required to navigate through extensive documents, sometimes exceeding 250 pages, to find pertinent details. This process was cumbersome and time-consuming, resulting in frustration and dissatisfaction among customers who sought quick resolution to their queries.

Moreover, the reliance on insurance advisors for clarifying doubts or obtaining information added another layer of complexity to the process. Customers had to wait for assistance, leading to delays in addressing their concerns and potentially impacting their overall experience with the insurance company.



Figure: 2.1.1: manual process

2.2 Proposed System

The proposed smart insurance processing system leverages UiPath automation and OpenAI natural language processing to streamline insurance workflows dramatically. Initially, the system automates the extraction of essential details from personal certificates, a task previously done manually by insurance agents. This automation not only reduces the time required for data entry but also minimizes the risk of errors, ensuring accurate and up-to-date customer information.

Furthermore, the system dynamically populates insurance forms with the extracted data, eliminating the need for agents to manually input this information into the insurance portal. This automation significantly reduces the turnaround time for processing insurance applications, enabling faster and more efficient service delivery to customers.

Additionally, the integration of OpenAI's natural language processing capabilities revolutionizes the way customer queries are handled. Instead of customers having to navigate through lengthy documents or rely on insurance advisors for assistance, they can now simply pose their queries to the AI system.

Overall, the proposed system not only streamlines insurance workflows but also enhances the customer experience. By automating manual tasks and improving query handling processes, the system reduces the burden on insurance agents and empowers customers with quicker access to information.



Figure: 2.2.1: Automatic process

2.3 Sequence / Flow Diagram

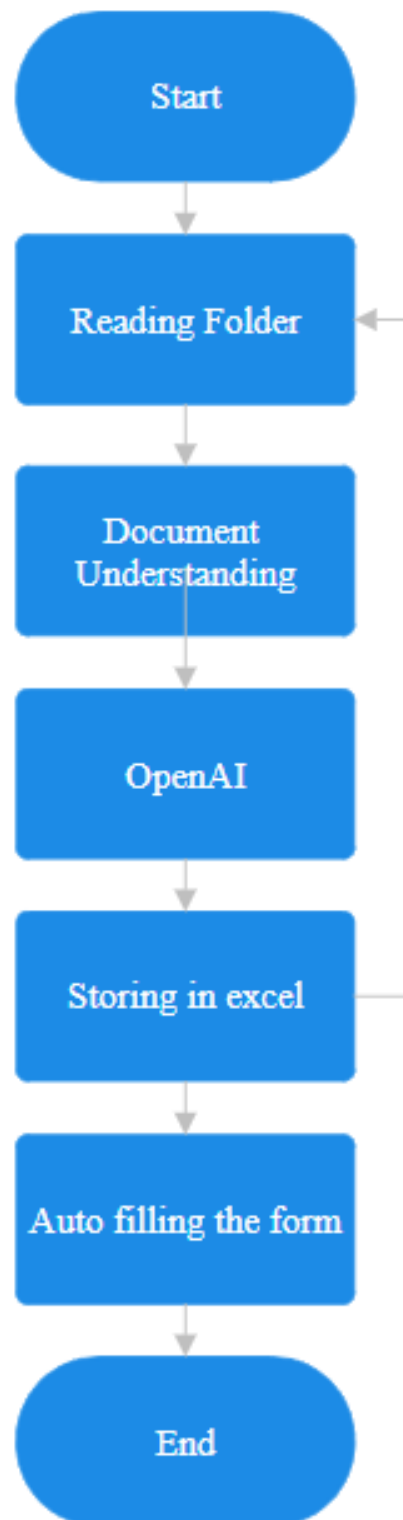


Figure: 2.3: Flow Chart

3. SYSTEM DESIGN

3.1 Input Design

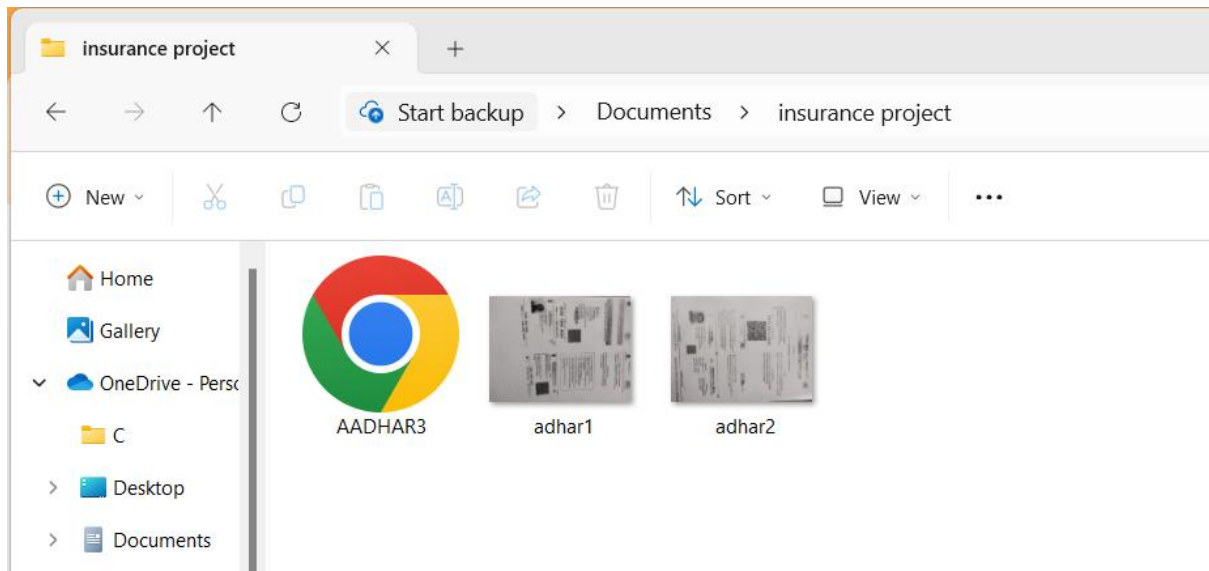


Figure: 3.1.1: Folder that contain personal certificates

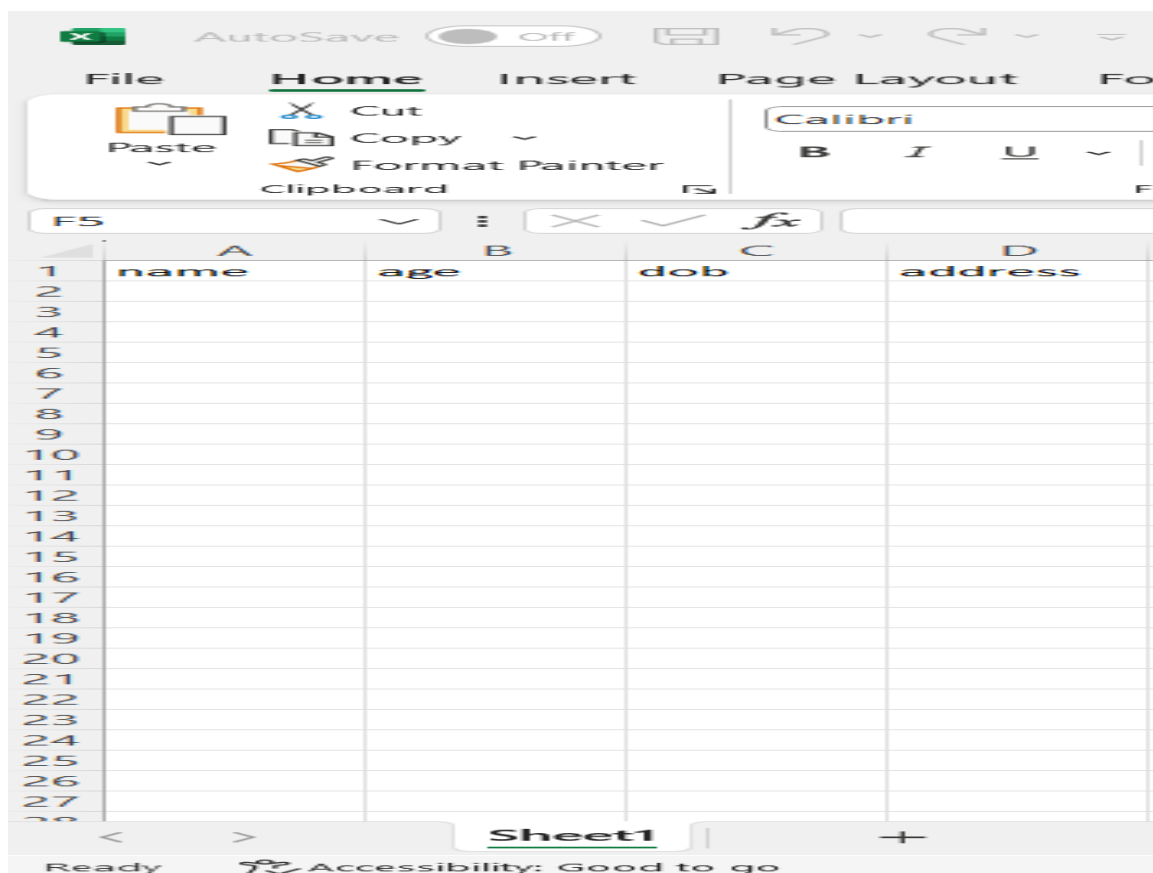
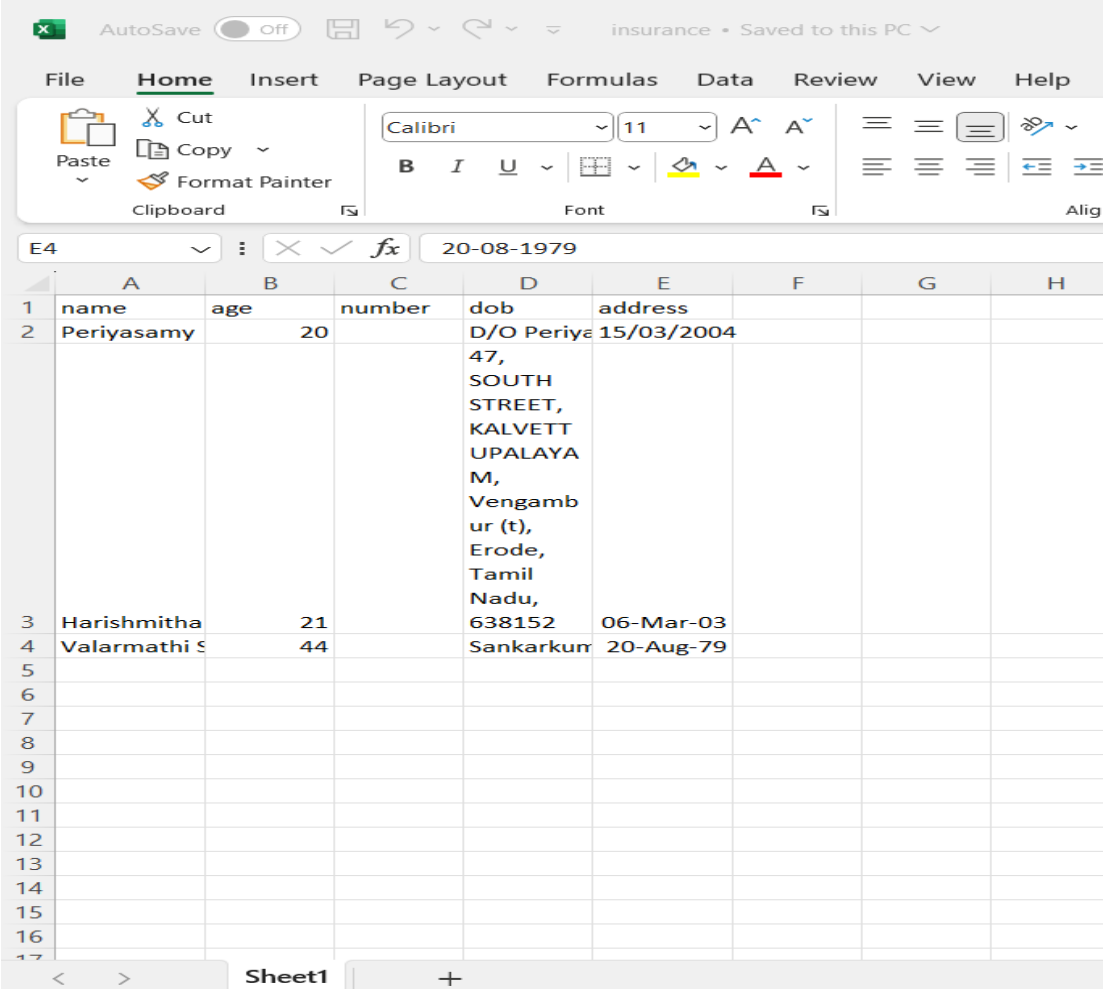


Figure: 3.1.2: Blank excel sheet

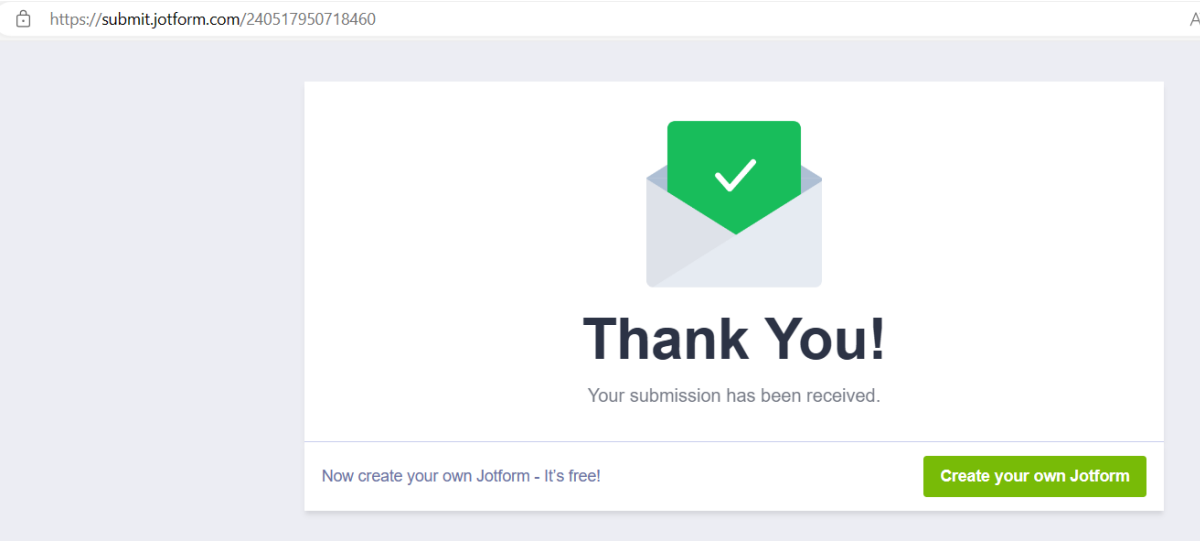
3.2 Output Design



The screenshot shows a Microsoft Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H
1	name	age	number	dob	address			
2	Periyasamy	20		D/O Periya 47, SOUTH STREET, KALVETT UPALAYA M, Vengamb ur (t), Erode, Tamil Nadu, 638152	15/03/2004			
3	Harishmitha	21		638152	06-Mar-03			
4	Valarmathi S	44		Sankarkun	20-Aug-79			
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								

Figure: 3.2.1: Filled excel



The screenshot shows a web browser window with the URL <https://submit.jotform.com/240517950718460>. The page displays a confirmation message:

Thank You!
Your submission has been received.

Now create your own Jotform - It's free!

[Create your own Jotform](#)

Figure: 3.2.2: Auto filled form

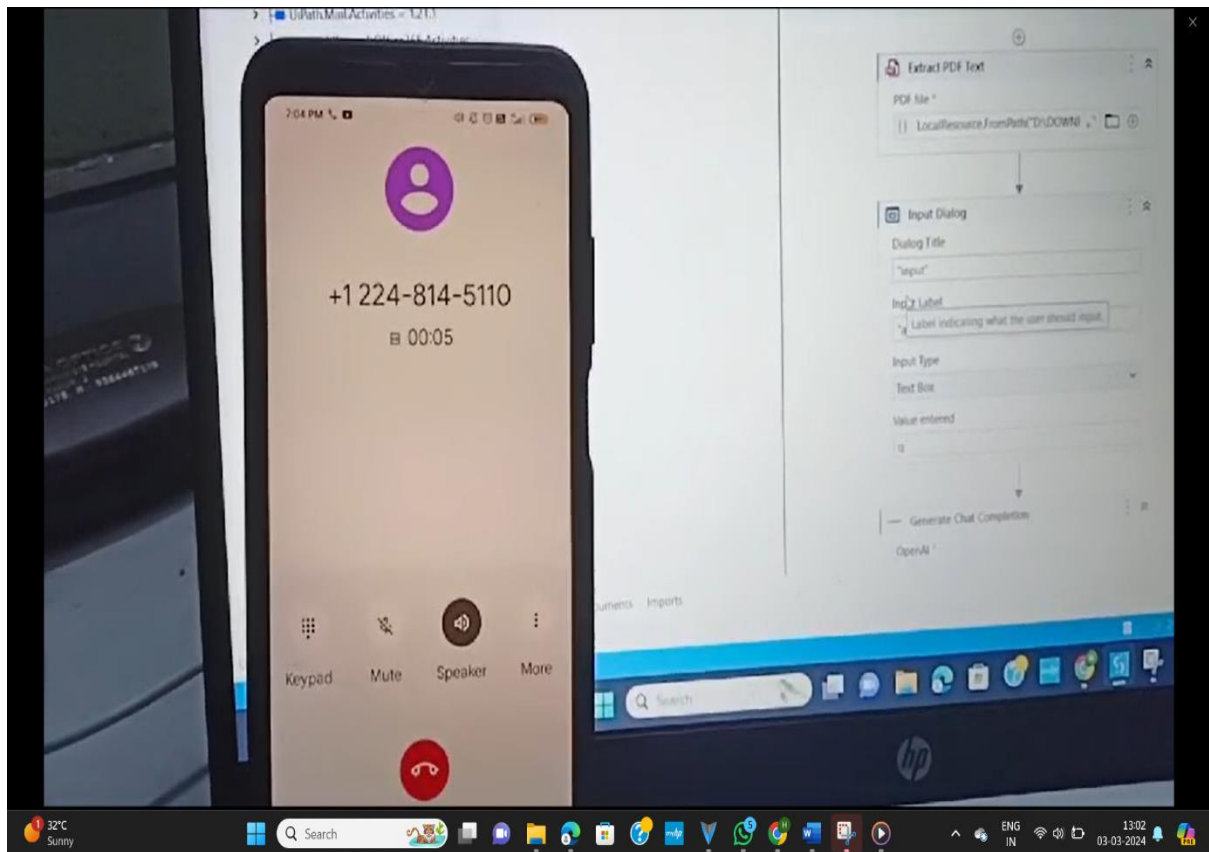


Figure: 3.2.3: Automatic call

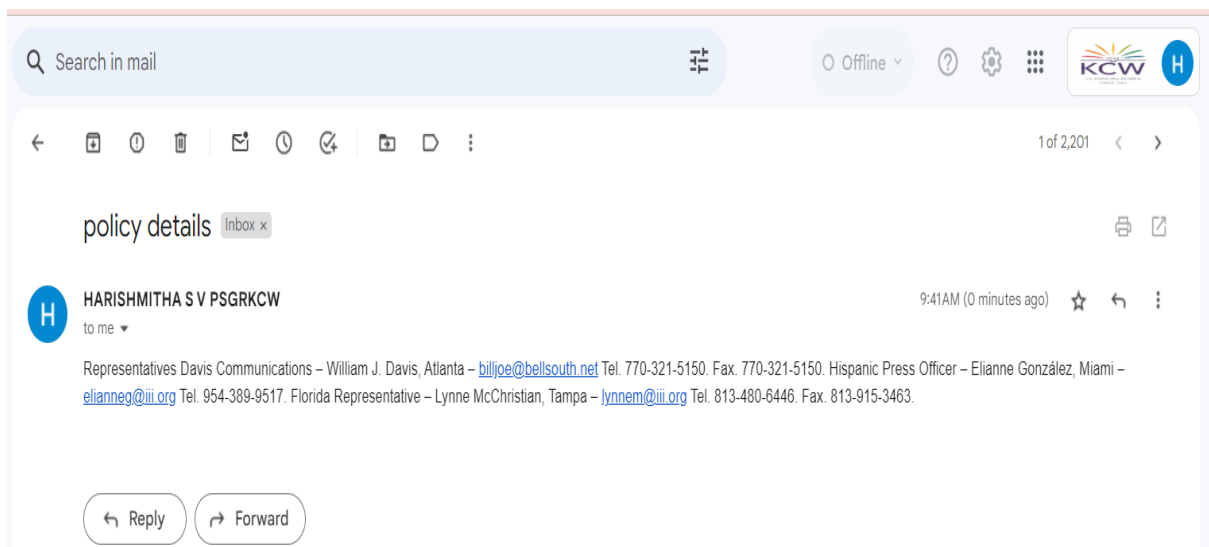


Figure: 3.2.3: Automatic email

4. SYSTEM IMPLEMENTATION

4.1 RPA Implementation Process

Implementing a project like "Smart Insurance Processing" with UiPath and OpenAI involves several steps. Here's a high-level implementation process:

1. Requirement Gathering:

- Understand the requirements of the project thoroughly.
- Identify the sources from which personal details need to be fetched (e.g., personal certificates).
- Define the insurance form format and fields that need to be filled.
- Determine the modes of communication for answering queries (message, email, phone call).

2. Setting up UiPath Environment:

- Install UiPath Studio and UiPath Robot on the development machine.
- Set up necessary dependencies or libraries for interacting with Excel, email, and other systems.

3. Data Extraction and Storage:

- Develop a workflow in UiPath for fetching personal details from personal certificates.
- Extract relevant data (e.g., name, address, insurance details) and store it in an Excel spreadsheet or a database.

4. Automated Form Filling:

- Create a UiPath workflow to automatically fill in the insurance form using the data stored in the Excel spreadsheet.
- Ensure accuracy and validation checks during the form filling process.

5. Integrating OpenAI for Query Handling:

- Utilize OpenAI's API to integrate natural language understanding capabilities.
- Develop a module in UiPath to send queries to OpenAI and receive responses.
- Design prompts for users to ask queries related to insurance.

6. Mode of Communication:

- Implement logic to allow users to choose their preferred mode of communication (message, email, phone call).
- Set up integration with messaging platforms (e.g., Twilio for SMS), email services, and telephony APIs.

7. Handling Queries:

- Process user queries received through the chosen mode of communication.
- Use natural language processing to understand and interpret user queries.
- Send queries to OpenAI and retrieve responses.
- Format and deliver responses through the selected communication channel.

8. Testing and Validation:

- Test the entire workflow thoroughly to ensure accuracy, reliability, and security.
- Validate the extracted data, form filling process, and query handling mechanisms.
- Conduct user acceptance testing to gather feedback and make necessary improvements.

9. Deployment:

- Deploy the solution in the production environment.
- Monitor the system for any issues and perform necessary maintenance.
- Provide training to end-users on how to interact with the system effectively.

10. Maintenance and Updates:

- Regularly update the system to incorporate improvements and address any issues.
- Stay updated with changes in regulations, insurance forms, and communication channels.
- Continuously optimize the workflow for better performance and efficiency.

By following these steps, we can successfully implement the "Smart Insurance Processing" project using UiPath and OpenAI

4.2 Results of Execution

project on smart insurance processing represents a significant advancement in streamlining insurance operations through automation and artificial intelligence integration. By leveraging UiPath's robotic process automation capabilities, you've effectively automated the extraction of personal details from certificates, eliminating tedious manual tasks and ensuring accuracy in data capture. This foundational step lays the groundwork for efficient insurance processing by providing a reliable source of information for further analysis and action.

With the extracted data seamlessly stored in an Excel sheet, your system facilitates easy access and management of vital information. This structured approach enhances data organization and simplifies subsequent processes such as form filling. Through UiPath's automation features, you've successfully automated the population of insurance forms with the data extracted from personal certificates. This not only reduces the likelihood of errors inherent in manual data entry but also accelerates the processing time, leading to faster insurance approvals and customer satisfaction.

Integration with OpenAI marks a significant milestone in enhancing the intelligence of your insurance processing system. By leveraging OpenAI's natural language processing capabilities, your system can interpret user queries related to insurance documents effectively. This empowers users to seek clarification or information about their insurance policies in a seamless and intuitive manner. Furthermore, the flexibility to choose preferred modes of communication, such as messages, emails, or phone calls, ensures that responses are delivered through channels that best suit the user's convenience and preferences.

The synergy between automation technologies like UiPath and advanced AI from OpenAI not only optimizes insurance processing workflows but also lays the foundation for scalable and adaptable systems. The modular design of your project allows for easy integration with existing processes and potential expansion to accommodate evolving needs



Figure: 4.2.1:
Extracting details



Figure: 4.2.2:
Form Auto filling



Figure: 4.2.3:
Policy Details



Figure: 4.2.4:
Phone call

5. CONCLUSION

The "Smart Insurance Processing" project marks a significant milestone in revolutionizing insurance operations through the amalgamation of UiPath and OpenAI technologies. This innovative fusion streamlines the traditionally labour-intensive processes of data extraction, form filling, and customer support. By automating the extraction of personal data from certificates, the project eliminates errors and accelerates the insurance application process. This not only enhances efficiency but also mitigates potential bottlenecks, allowing insurance providers to serve their customers with greater agility and responsiveness.

Moreover, the seamless integration between UiPath and OpenAI facilitates a smooth transition of data from extraction to form completion. Through intelligent automation, the extracted data seamlessly populates insurance forms, sparing customers the burden of manual input and ensuring accuracy throughout the process. This integration not only enhances productivity but also optimizes resource allocation within insurance organizations, freeing up personnel to focus on more strategic tasks and value-added activities.

Furthermore, the project prioritizes customer-centricity by offering a personalized and flexible communication framework. By enabling users to choose their preferred mode of interaction—be it messages, emails, or phone calls—the project ensures that customer queries are addressed promptly and effectively. This customizable approach not only enhances the overall customer experience but also fosters greater engagement and satisfaction, thereby strengthening the relationship between insurers and policyholders.

Additionally, the incorporation of AI-driven support for customer queries adds another layer of sophistication to the project. By leveraging OpenAI's advanced natural language processing capabilities, the system can intelligently interpret and respond to user inquiries, providing timely and accurate assistance. Whether through automated messages, emails, or phone calls, customers receive the support they need in a manner that aligns with their preferences, further enhancing the overall efficiency and effectiveness of the insurance process.

In conclusion, the "Smart Insurance Processing" project represents a paradigm shift in insurance operations, harnessing the power of automation and artificial intelligence to drive efficiency, accuracy, and customer satisfaction. By seamlessly integrating UiPath and OpenAI technologies, the project not only optimizes internal processes but also elevates the customer experience to new heights.

5.1 Scope of future

The project on smart insurance processing, leveraging UiPath and OpenAI, represents a transformative approach to streamlining insurance workflows. By harnessing the power of automation and artificial intelligence, the system excels in extracting pertinent details from personal certificates, effortlessly populating insurance forms, and providing personalized assistance to users. This amalgamation of technologies marks a significant advancement in the insurance sector, offering unparalleled efficiency and convenience.

One key aspect of future development lies in refining the data extraction process. Incorporating advanced OCR techniques and machine learning algorithms can enhance accuracy and adaptability to various certificate formats, ensuring seamless data retrieval. This evolution will pave the way for more reliable and comprehensive processing of insurance-related documents, thereby boosting overall efficiency and user satisfaction.

Moreover, the integration of sophisticated Natural Language Understanding (NLU) models presents a promising avenue for improving user interactions. By training the AI model on a broader spectrum of insurance queries and responses, the system can deliver more nuanced and contextually relevant answers, fostering enhanced user engagement and comprehension. This advancement underscores the project's commitment to continuously enhancing user experiences through cutting-edge AI technologies.

Additionally, the incorporation of real-time communication channels, such as chatbots and mobile apps, will revolutionize user support and engagement. By providing instant assistance and personalized recommendations through these channels, the project can significantly enhance customer satisfaction and loyalty. This proactive approach to customer service underscores the project's commitment to leveraging technology for fostering stronger customer relationships and driving business growth.

In essence, the scope of future development for the smart insurance processing project is vast and multifaceted. From enhancing data extraction accuracy to expanding integration with diverse systems and platforms, the project holds immense potential for reshaping the insurance landscape through automation and AI-driven innovation. By embracing these opportunities for growth and evolution, the project can continue to deliver tangible benefits to insurance companies and users alike, propelling the industry toward a more efficient and customer-centric future.

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- UiPath Activities for Phone Calls:
<https://docs.uipath.com/activities/other/latest/legacy-integrations/amazoncomprehend-invoke-method>